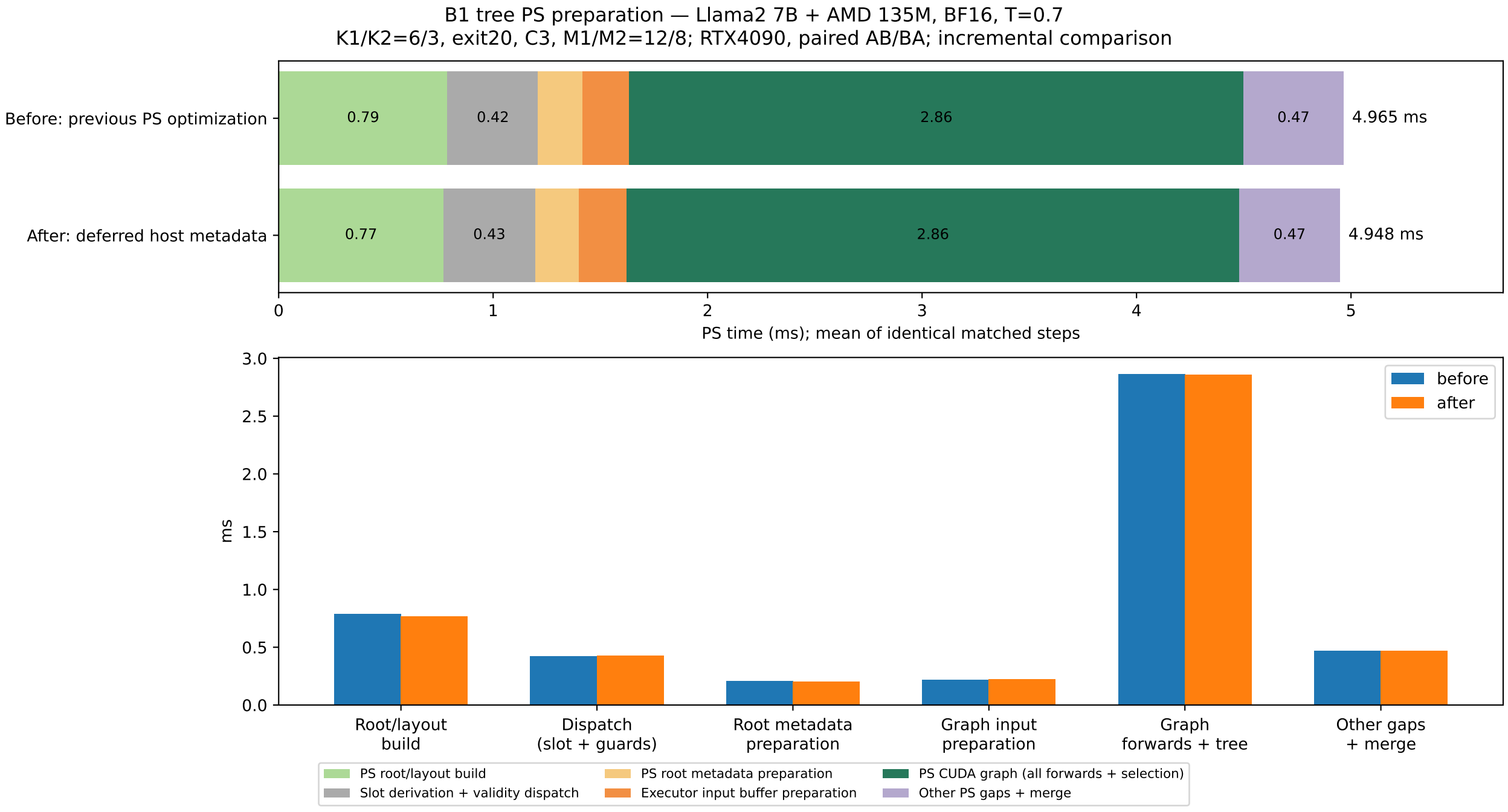
B1 tree PS preparation — Llama2 7B + AMD 135M, BF16, T=0.7
K1/K2=6/3, exit20, C3, M1/M2=12/8; RTX4090, paired AB/BA; incremental comparison
Before: previous PS optimization
0.79
0.42
2.86
0.47
4.965 ms
After: deferred host metadata
0.77
0.43
2.86
0.47
4.948 ms
0
1
2
3
4
5
PS time (ms); mean of identical matched steps
3.0
2.5
2.0
1.5
1.0
0.5
0.0
ms
before
after
Root/layout build
Dispatch (slot + guards)
Root metadata preparation
Graph input preparation
Graph forwards + tree
Other gaps + merge
PS root/layout build
Slot derivation + validity dispatch
PS root metadata preparation
Executor input buffer preparation
PS CUDA graph (all forwards + selection)
Other PS gaps + merge